

Seeing Green from Space: Machine Learning for Greenspace Detection

Martina Pardy
University of Liverpool
London School of Economics

20th CGA Annual Conference
Boston

October 2-3, 2026



Workshop outline

This is an **introductory workshop**. It assumes no prior knowledge of satellite imagery or machine learning.

1. Introduction to Satellite Data (15 mins)

applied in R

- ▶ Multispectral satellite instruments and images
- ▶ Key spectral indices, loading and visualising imagery

2. Machine Learning for Spatial Classification (15 mins)

theory

- ▶ Supervised vs. unsupervised ML for land use classification
- ▶ Clustering and Random Forest

3. Applied Land Use Classification

applied in R

- ▶ Classifying imagery into greenspace

Setting up the applied part

Choose how you would like to follow the applied part of the workshop:

- ▶ Follow along without coding
- ▶ **Run the exercises online via Binder**

Open the GitHub repository:

<https://github.com/mpardy/satellite-imagery-and-machine-learning>

Built on:

- ▶ LSE GY476, *Applied Geographic Information Systems*
- ▶ NASA ARSET, https://github.com/NASAARSET/LCLUC_2026
- ▶ Classifying satellite imagery in R,
https://urbanspatial.github.io/classifying_satellite_imagery_in_R/

Imago: Make (satellite) imagery more useful, usable, and used

Imagery Innovation



Ingesting pixels,
producing
metrics at scale

Data for all



Meeting users
where they are

Capability & Community



Building a
community
around imagery

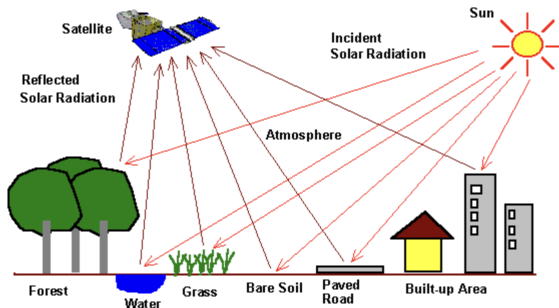


imago.ac.uk

Part 1: Introduction to Satellite Data

Optical Remote Sensing

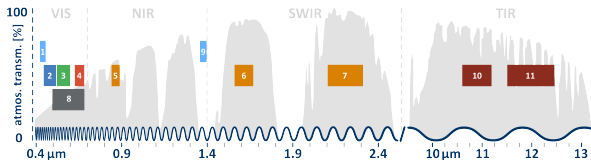
Optical sensors rely on the sun. After the atmosphere filters it, useful energy remains only within roughly 0.25–3 micrometres (μm).



Source: CRISP, National University of Singapore, *Principles of Remote Sensing* (crisp.nus.edu.sg). © CRISP 2001; curves redrawn.

What is a satellite image?

- ▶ Most products measure **reflectance**: the intensity of the sun's radiation reflected by the Earth's surface
- ▶ Reflectance is recorded across **different wavelengths** of the electromagnetic spectrum
- ▶ Because each place is measured at several wavelengths, one image holds multiple values per pixel, stored as separate raster layers
- ▶ These layers are called **bands**; the grid cells are **pixels**



Source: USGS; blogs.fu-berlin.de/reseda/landsat-8

Example: Landsat 8/9

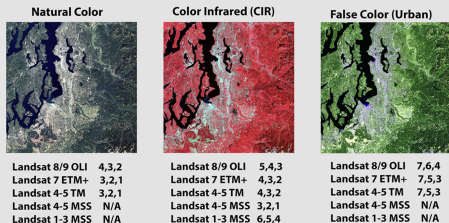
Band	Name	Wavelength (μm)	Resolution (m)
1	aerosols	0.44–0.45	30
2	blue	0.45–0.51	30
3	green	0.53–0.59	30
4	red	0.64–0.67	30
5	NIR	0.85–0.88	30
6	SWIR-1	1.57–1.65	30
7	SWIR-2	2.11–2.29	30
8	pan	0.50–0.68	15
9	cirrus	1.36–1.38	30
10	TIR-1	10.6–11.2	100
11	TIR-2	11.5–12.5	100

The same pixels combine into different **composites**:

- ▶ **Natural colour** (4,3,2): close to what the eye sees
- ▶ **Colour infrared** (5,4,3): vegetation reflects NIR, so red marks healthy plants; water dark, urban white
- ▶ **False colour, urban** (7,6,4): vegetation in greens (darker = denser); urban blue, soils brown

Common Landsat Band Combinations

Images: Landsat 8 Path 46 Row 27 acquired August 23, 2020. Band numbers displayed as R,G,B.



Source: USGS Landsat 8–9 band specifications and band-combination composites (usgs.gov)

Different resolutions

Four kinds of **resolution** decide what we can observe:

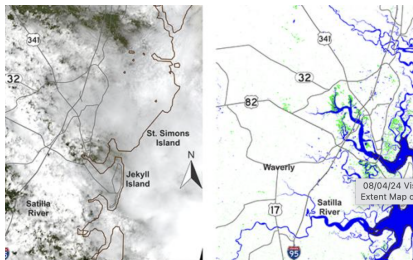
- ▶ **Spatial**: ground size of each pixel (level of detail)
- ▶ **Temporal**: how often a location is revisited
- ▶ **Spectral**: number and width of wavelength bands
- ▶ **Radiometric**: sensitivity to brightness differences

Applications across social science

Satellite imagery now supports research and policy across many domains:

- ▶ **Land and environment:** land-use change, deforestation
- ▶ **Urban:** settlement mapping, urban expansion, urban heat
- ▶ **Disasters:** floods and wildfires (next slides)

Application: near real-time flood mapping



Surface water (blue) and flood water (red).

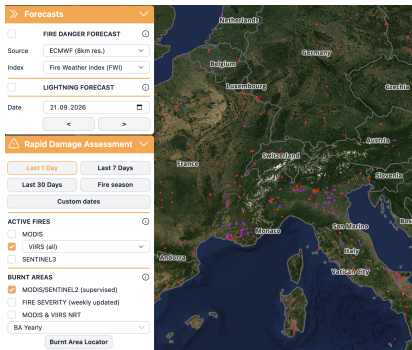
Credit: NASA Worldview

NASA Global Flood Product

- ▶ Near real-time flood maps at ~ 250 m, within 3 hours of observation
- ▶ Built from MODIS and VIIRS sensors, delivered through NASA **LANCE**
- ▶ Used for disaster response and flood-risk assessment

earthdata.nasa.gov/topics/human-dimensions/floods

Application: wildfire monitoring



Copernicus EFFIS, Current Situation
Viewer

European Forest Fire Information System

- ▶ Active fires updated 6 times daily, within 2–3 hours of acquisition
- ▶ Burnt-area maps from Sentinel-2, MODIS, and VIIRS imagery
- ▶ Fire-danger forecasts up to 6 days ahead

forest-fire.emergency.copernicus.eu

Spectral indices

A **spectral index** combines bands into a single number that isolates a surface property. The most common is the **Normalised Difference Vegetation Index (NDVI)**:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

Healthy vegetation reflects strongly in the near-infrared and absorbs red, so it scores high (roughly 0.2 to 0.9); water and built-up land score low or negative. Values run from -1 to 1 .

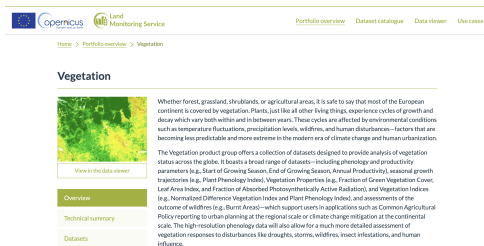
Other indices swap in different bands to target other surfaces:

- ▶ **Normalised Difference Water Index (NDWI)**:
(Green $-$ NIR) / (Green $+$ NIR)
- ▶ **Normalised Difference Built-up Index (NDBI)**:
(SWIR $-$ NIR) / (SWIR $+$ NIR)

Existing vegetation datasets

Ready-made vegetation products you can use without building your own:

Copernicus Land



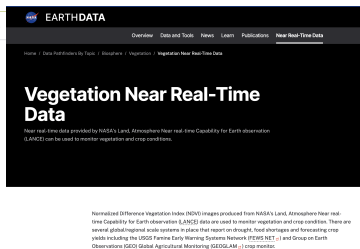
The screenshot shows the Copernicus Land Monitoring Service website. The header includes the European Union flag, the Copernicus logo, and the Land Monitoring Service logo. Navigation links include Portfolio overview, Dataset catalogue, Data viewer, and Use cases. The main content area is titled 'Vegetation' and features a satellite image of a forest. Below the image is a 'View in the data viewer' button. To the right of the image is a text block explaining that vegetation covers most of the European continent and is affected by environmental conditions like temperature, precipitation, and wildfires. Below this is a paragraph about the Vegetation product group, which offers a collection of datasets for analysis, including phenology and productivity parameters. A sidebar on the left contains links for Overview, Technical summary, and Datasets.

Vegetation indices and biophysical variables (NDVI, LAI, FAPAR)

[https:](https://land.copernicus.eu/en/products/vegetation)

[//land.copernicus.eu/en/products/vegetation](https://land.copernicus.eu/en/products/vegetation)

NASA Earthdata



The screenshot shows the NASA Earthdata website. The header includes the NASA logo and the Earthdata logo. Navigation links include Overview, Data and Tools, News, Learn, Publications, and Near Real-Time Data. The main content area is titled 'Vegetation Near Real-Time Data' and features a large heading. Below the heading is a paragraph explaining that near real-time data is provided by NASA's Land, Atmosphere Near real-time Capability for Earth observation (LANCE) and can be used to monitor vegetation and crop conditions. At the bottom, there is a paragraph about the Normalized Difference Vegetation Index (NDVI) images produced from NASA's Land, Atmosphere Near real-time Capability for Earth observation (LANCE) data, which are used to monitor vegetation and crop conditions. There are several global/regional scale systems in place that report on drought, food shortages and forecasting crop yields including the US25 Famine Early Warning Systems Network (FEWS NET) and Group on Earth Observations (GEO) Global Agricultural Monitoring (GEOGLAM) crop monitor.

Near real-time vegetation indices (NDVI, EVI) from MODIS and VIIRS

<https://www.earthdata.nasa.gov/topics/biosphere/>

[vegetation/near-real-time-data](https://www.earthdata.nasa.gov/topics/biosphere/vegetation/near-real-time-data)

Part 2: Introduction to Machine Learning

What is machine learning?

Machine learning (a type of artificial intelligence) lets software applications to become more accurate at predicting outcomes without being explicitly programmed

The shift is in what the programmer supplies:

- ▶ **Traditional programming:** rules + data $\rightarrow$ output
- ▶ **Machine learning:** data + example outcomes $\rightarrow$ the rules (a **model**)

The model learns patterns (from examples), then applies them to new, unseen data

In this workshop: the model learns to label each pixel as landuse categories using supervised and unsupervised models

How does the model “learn”?

Unsupervised learning — no labels

- ▶ Finds patterns and structure within the data itself
- ▶ Tasks: clustering, dimensionality reduction, association rules

Supervised learning — labelled data

- ▶ Input data paired with known output labels
- ▶ Learns an input $\rightarrow$ output mapping to predict on unseen data
- ▶ Tasks: **classification** (categorical), **regression** (numerical)

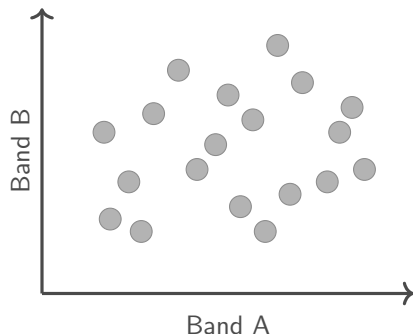
Either way, the model runs in two steps: **learning** (fit on training data) then **prediction** (apply to new data).

Part 2A: Unsupervised Machine Learning

K-means clustering

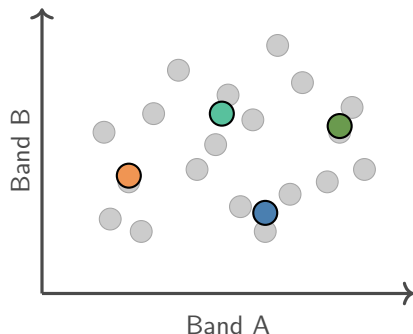
K-means clustering (simplified)

- ▶ Each pixel is projected into **feature space**
- ▶ Every dimension (band) is a measurement (reflectance)
 - ▶ Usually many dimensions; 2 shown here



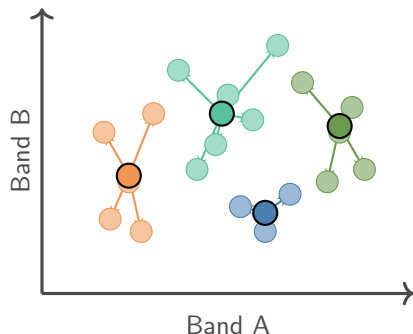
K-means clustering (simplified)

- ▶ Each pixel is projected into **feature space**
- ▶ Every dimension (band) is a measurement (reflectance)
 - ▶ Usually many dimensions; 2 shown here
- ▶ Pick **K random points** as class centres



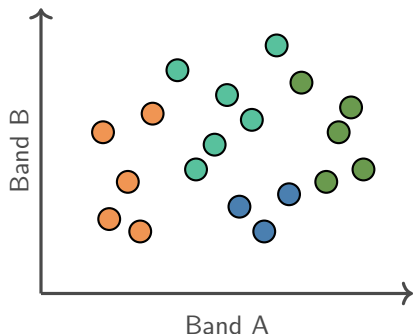
K-means clustering (simplified)

- ▶ Each pixel is projected into **feature space**
- ▶ Every dimension (band) is a measurement (reflectance)
 - ▶ Usually many dimensions; 2 shown here
- ▶ Pick **K random points** as class centres
- ▶ Group each point by distance to the closest centre



K-means clustering (simplified)

- ▶ Each pixel is projected into **feature space**
- ▶ Every dimension (band) is a measurement (reflectance)
 - ▶ Usually many dimensions; 2 shown here
- ▶ Pick **K random points** as class centres
- ▶ Group each point by distance to the closest centre
- ▶ Assign a class to each point from its nearest centre



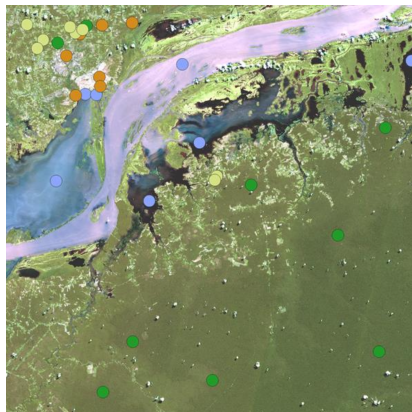
Part 2B: Supervised Machine Learning

K-means clustering

Classification: training data

Training data can be:

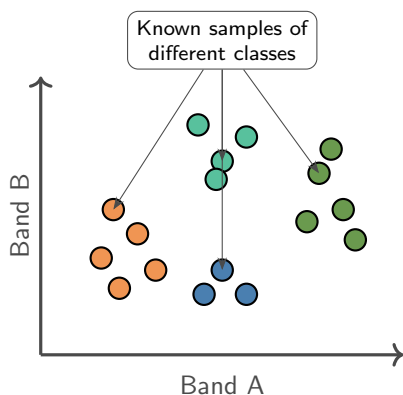
- ▶ Points, lines, or polygons
- ▶ The pixel value(s) under each feature give a **spectral profile**
- ▶ This profile is how much light is reflected in the wavelengths of each band



NASA ARSET: Land Cover and Land Use Change

Supervised classification: KNN overview

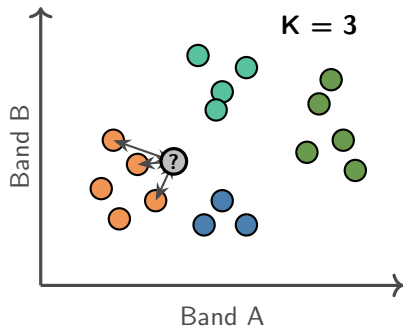
- ▶ **K-nearest neighbour** uses the labels of known samples (training data) to classify new ones
- ▶ K-means, by contrast, has no predefined classes; it only looks for similarity
- ▶ How it works:
 - ▶ For a new sample, check its K nearest known points
 - ▶ K is an integer > 1 (odd avoids ties)
 - ▶ The majority vote wins



Supervised classification: KNN step-by-step

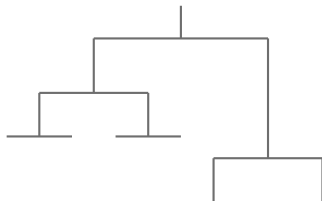
Watch how KNN classifies a new point:

- ▶ Set $K = 3$
- ▶ Find the 3 closest known points
- ▶ Tally the votes
- ▶ Assign the new point to the winning class



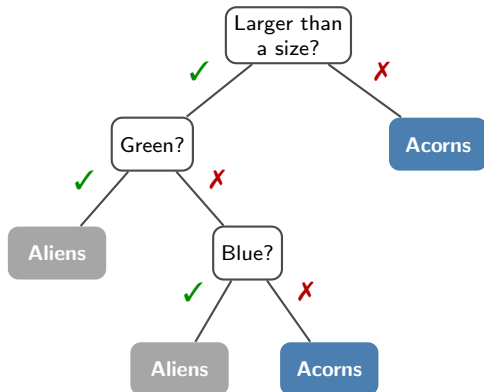
Random forest: overview

- ▶ Random forest is a **tree-based** model
- ▶ It builds an array of decision trees
- ▶ Each tree offers a possible way to split the training data
- ▶ New data is classified by applying the tree(s)



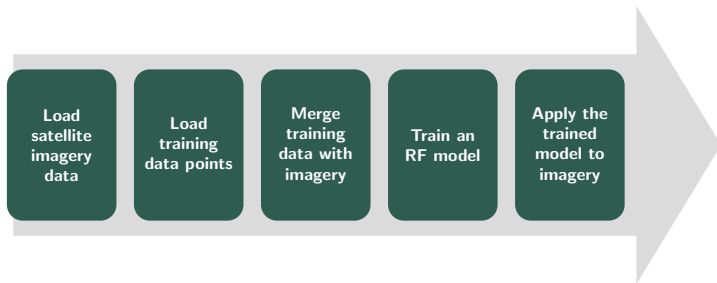
The decision tree: acorns or aliens?

- ▶ Each split is a binary (true/false) question
- ▶ “Larger than some size?”
- ▶ “Is it green?”
- ▶ “Is it blue?”
- ▶ Aliens can be blue or green
- ▶ Acorns are smaller than aliens



Building a random forest model in R

Roadmap for the following demo



Where is the frontier?

Today we classify each **pixel** from its spectral bands, treating pixels independently. The field is moving beyond that:

Deep learning and segmentation

- ▶ Convolutional networks (e.g. U-Net) use spatial context, not just one pixel's spectrum: texture, shape, and neighbourhood
- ▶ This turns per-pixel classification into **semantic segmentation**, the current state of the art for land cover

Foundation models and embeddings

- ▶ Models pre-trained on vast unlabelled imagery turn each pixel into a rich **embedding**: a vector capturing learned patterns

Thank you

m.l.pardy@liverpool.ac.uk · m.l.pardy@lse.ac.uk

Imago: Data Service for Imagery
Geographic Data Science Lab, University of Liverpool



imago.ac.uk



Appendix

Sources

Data and tools

- ▶ Landsat imagery: USGS EarthExplorer, <https://earthexplorer.usgs.gov>
- ▶ Sentinel imagery: Copernicus, <https://dataspace.copernicus.eu>
- ▶ Google Earth Engine: Gorelick et al. (2017), *Remote Sensing of Environment*
- ▶ Flood mapping: NASA LANCE, <https://www.earthdata.nasa.gov/topics/human-dimensions/floods>
- ▶ Wildfire monitoring: Copernicus EFFIS, <https://forest-fire.emergency.copernicus.eu>

Figures and adapted material

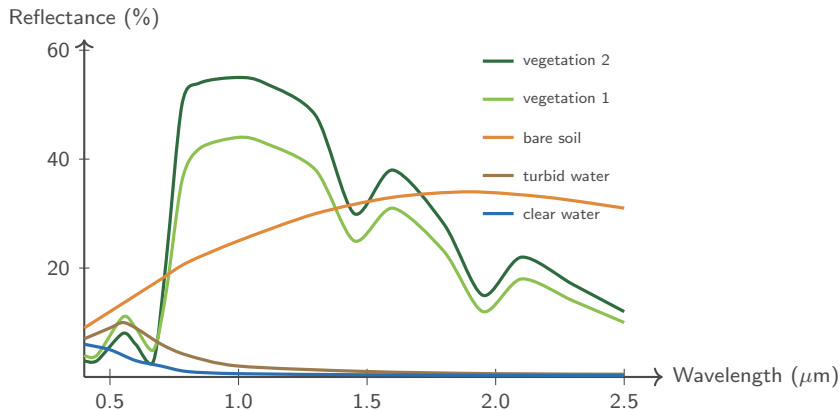
- ▶ Landsat band designations and composites: USGS, <https://www.usgs.gov/landsat-missions/landsat-8>
- ▶ Electromagnetic spectrum figure: FU Berlin RESEDA, <https://blogs.fu-berlin.de/reseda/landsat-8>
- ▶ KNN, K-means, and random forest diagrams: adapted from NASA ARSET, *Visualizing Land Cover and Land Use Change with NASA Satellite Imagery*
- ▶ Segmentation vs classification figure: <https://medium.com/inovako/object-detection-vs-image-segmentation-e5290e4690d>

Statistics

- ▶ Earth observation satellite growth and data volumes: Wilkinson et al. (2024), *Science of the Total Environment* 909, 168584

Spectral reflectance signatures

Different surfaces reflect and absorb differently by wavelength, giving each a **signature** that lets us tell them apart.



Source: CRISP, National University of Singapore, *Principles of Remote Sensing* (crisp.nus.edu.sg). © CRISP 2001; curves redrawn.

Training data quality

Good training data is:

- ▶ High-quality (accurate)
- ▶ Representative (varied)

A model trained on good data:

- ▶ Is robust to variation in the land cover itself (e.g. water turbidity)
- ▶ Is robust to errors between the land cover and the sensor (e.g. atmospheric effects)

K-means clustering

Measuring distance in K-means

Each pixel is a vector across its bands, and K-means assigns it to the nearest centre. The standard metric is **Euclidean distance**:

$$d(\mathbf{x}, \mathbf{c}) = \sqrt{\sum_{b=1}^p (x_b - c_b)^2}$$

Other metrics count distance differently:

- ▶ **Manhattan**: sums differences band by band; less swayed by outliers
- ▶ **Mahalanobis**: accounts for correlation between bands
- ▶ **Cosine**: compares spectral shape, not brightness

Standard K-means assumes Euclidean distance

What K-means minimises

K-means places centres to minimise the **within-cluster sum of squares**:

$$J = \sum_{k=1}^K \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \mathbf{c}_k\|^2$$

Dividing by the number of points gives the **within-cluster variance**, so K-means minimises the spread inside clusters.

It alternates two steps until assignments stop changing:

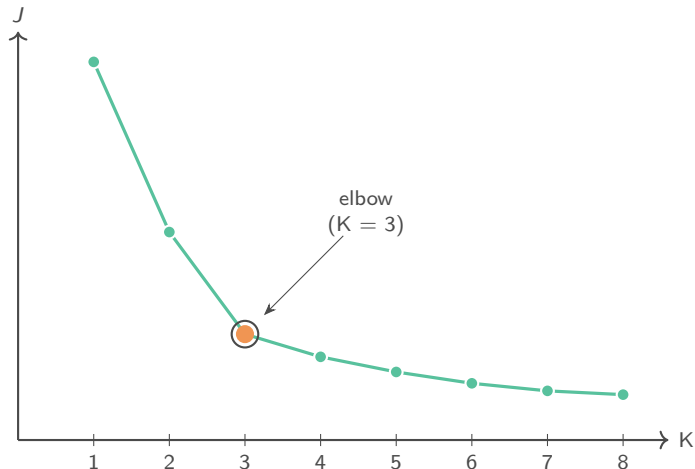
- ▶ Assign each point to its nearest centre
- ▶ Move each centre to the **mean** of its points

J falls at every step, converging to a local optimum, so the random start matters.

Choosing K: J always falls as K grows; the “elbow” is a common heuristic

Choosing K: the elbow

Run K-means for a range of K and plot J (within-cluster sum of squares) against K . The **elbow** is where the curve stops dropping steeply.



Past the elbow, extra clusters cut J only slightly.

Decision tree: how splits are chosen

- ▶ At each node, the tree evaluates all candidate satellite bands and reflectance thresholds to find the split that best separates the land cover classes (lowest impurity)
- ▶ For continuous reflectance data (e.g. Band 5 or NDVI), it tests cutoffs between observed values: the exact threshold (e.g. $\text{Band 5} < 15,337$) is learned automatically from the data, not set by hand
- ▶ The algorithm recursively splits each branch until nodes are mostly pure or a maximum depth is reached

In `rpart`, this creates the readable flowchart tree diagram we plot in R.

Choosing the best model

No single model wins everywhere, so we compare candidates (e.g. decision trees, KNN, random forest) on the **same held-out data**:

- ▶ **Split** the data: train on one part, test on another that the model never saw
- ▶ **Cross-validate**: rotate the split several times so the score does not depend on one lucky partition
- ▶ **Score** each model on a metric that matches your objective
- ▶ **Select**, then report performance on a final untouched test set

The confusion matrix is the foundation for most classification metrics, so we start there.

The confusion matrix

Count predictions against the truth. For greenspace detection:

		Actual	
		Greenspace	Not greenspace
Predicted	Greenspace	True positive 180	False positive 20
	Not greenspace	False negative 40	True negative 760

Diagonal (teal) = correct; off-diagonal (orange) = errors. The two error types matter differently: a **false negative** misses real greenspace, a **false positive** invents it.

Precision vs. recall: the trade-off

Precision

Of all pixels **predicted** as greenspace, how many truly are?

$$\text{Precision} = \frac{TP}{TP + FP}$$

Low precision → **over-prediction**: the map marks too much as green

Recall

Of all pixels that **truly are** greenspace, how many are detected?

$$\text{Recall} = \frac{TP}{TP + FN}$$

Low recall → **under-detection**: the map misses real greenspace

The trade-off: lowering the decision threshold catches more true greenspace (higher recall) but admits more false positives (lower precision). Tune the threshold on a validation set to balance the two for your task.

Foundation model embeddings

What is an embedding?

Instead of feeding raw pixel values (spectral bands) into a model, a **foundation model** first turns each pixel into a richer representation, the **embedding**: a vector of numbers capturing patterns the model learned from vast amounts of imagery.

Geospatial foundation models

Models such as GeoTessera are pre-trained on large-scale satellite imagery and convert each pixel into a fixed-length embedding (e.g. 64 or 128 dimensions) in place of its original bands.

Why embeddings help:

- ▶ They encode spatial and spectral patterns a single pixel's bands cannot
- ▶ Higher dimensions capture more detail, at higher compute cost
- ▶ Classifying on embeddings often needs less labelled data to reach good accuracy